

Extending the QSC Atlas: from where a country stands to how credible its approach is

A proposal for one evaluative view, grounded in the assessment literature

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Summary. The Atlas maps where each country stands in the quantum-safe transition: its coordination bloc, its standards role, and the instruments that bind it. It does not yet say how good that approach is. This note proposes a single addition, an evaluative layer that scores the credibility of each country's quantum-safe governance against defined criteria, drawn from the comparative-evaluation method that the assessment literature has converged on and adapted to the Atlas as a standing, auditable base. The aim is one justified view, not a second instrument: it reuses fields the Atlas already holds, adds a small and defensible criteria set, and is reported as provisional, consistent with the governance-of-expectations frame the Atlas rests on.

1. What the Atlas measures, and the question it leaves open

The Atlas answers a structural question well. For each country it records the coordination bloc it migrates with, its standards role, and the governing instrument, its legal status and the obligation it imposes. That is a map of position: where a country sits in the transition. Position is necessary, and it is what a first reader needs. It is not sufficient for the question a policy or advisory reader asks next, which is evaluative rather than descriptive: is this country's approach any good. Two countries can share a bloc, a role and a binding instrument, and differ sharply in whether their strategy is coherent with wider policy, whether it is measured, and whether it is governed by a body with the mandate and the skills to deliver it. The Atlas cannot currently show that difference.

2. The literature offers a method, and names the Atlas as its base

The QSA methods synthesis re-reads the academic and policy corpus for ways to assess a quantum-safe transition, and reaches a clear finding: no assessment framework specific to quantum governance yet exists, a point Paglieri et al. (2025) state directly and build around. Most of the methods the synthesis assembles are case-level, built to assess an organisation or a venture. One is national in scope and therefore fits the Atlas's unit of analysis: the comparative policy evaluation that Paglieri et al. (2025) construct from OECD and World Economic Forum practice. The synthesis notes, in passing, that the Atlas could provide the auditable base such an evaluation needs, with the added property that it is current. This proposal takes that observation as its starting point: port the one national method into the Atlas as a standing layer, so the Atlas becomes the evaluative base the literature says is missing.

3. The proposed view: a governance-quality layer

The addition is an evaluative reading of each country on six criteria, scored from the documents the Atlas already cites. Relevance asks whether the national approach is aligned with the threat and carries a stated transition plan. Coherence asks whether it is integrated with wider cyber and science-and-technology policy and was built with the state, industry and academia consulted. Effectiveness, judged by design rather than by outcome, asks whether milestones, measurement and an inventory duty exist. Efficiency asks whether the instrument mix is proportionate to the stated objective. Governance quality asks whether a mandate, a lead authority, a steering function and the skills to run the migration are in place. Impact, again by design, asks whether the approach is structured to reduce the specific exposure, under a window of risk rather than a single date. Table 1 sets out each criterion, its theoretical grounding, and the Atlas field it reads from.

Criterion	What it scores for a country	Theoretical grounding	Reads from
Relevance	Alignment with the threat; a stated transition plan to quantum-safe cryptography	Comparative evaluation, after OECD and NIST (Paglieri et al., 2025)	Main regulation; migration timeline
Coherence	Integration with wider cyber and S&T policy; breadth of consultation across state, industry and academia	OECD and WEF practice (Paglieri et al., 2025); the triple helix (Purohit et al., 2023)	Gov actors; standards participation
Effectiveness (by design)	Presence of milestones, measurement and a cryptographic-inventory duty, not yet outcomes	Process evaluation (Vance, 2025); CEPS recommendation on inventories (Pupillo et al., 2025)	Legal status; migration timeline; inventory mandate
Efficiency	Proportionality of the instrument mix to the stated objective	Comparative evaluation, after OECD (Paglieri et al., 2025)	Legal status; obligation
Governance quality	A mandate, a lead authority, a steering function and migration skills	WEF practice (Paglieri et al., 2025); CEPS capability recommendations (Pupillo et al., 2025)	Gov actors; capacity signals
Impact (structured-for)	Whether the approach is structured to reduce the specific exposure, under a Q-period not a Q-day	Impact evaluation (Vance, 2025); the move to a Q-period (Pupillo et al., 2023; 2025)	Target completion; standards role

Table 1. The evaluative layer: six criteria, their grounding, and the Atlas fields they read from.

4. Why this is the right view, and what keeps it honest

The criteria are not invented for the Atlas; they are the OECD and WEF evaluation criteria that Paglieri et al. (2025) adapted for national quantum strategies, which is why they transfer to a country instrument cleanly. Three further models sharpen the reading. Vance (2025) separates evaluation into process, impact and cost-benefit, which is why the layer judges effectiveness and impact by design rather than by outcomes an emerging field cannot yet show. Marchant et al. (2024) read emerging-technology governance as a movement from soft law to hard law, which gives the layer a way to read a country's trajectory, whether its instruments are hardening, rather than a fixed snapshot. The CEPS reports represent the timing variable as a

window of probability, a Q-period rather than a Q-day (Pupillo et al., 2023; 2025), so the impact criterion is scored against an honest distribution rather than a single date.

The whole layer sits inside the frame the Atlas already uses. Governance of expectations treats this transition as provisional, because the threat's timing is genuinely unknowable (Budde and Konrad, 2019). An evaluative layer built on that frame does not score a country as finished or correct; it scores how credible its approach is under irreducible uncertainty, and it expects the score to move. Two disciplines keep it defensible. Scoring is reconciled by a second reviewer before it is shown, the validation step Paglieri et al. (2025) build into their method. And the Atlas's existing confidence grade is carried onto the evaluation, so a country assessed on thin evidence reads as tentative rather than judged.

5. Why it does not overload the Atlas

The layer is one view, not a second instrument. It reads mostly from fields the Atlas already records, the regulation, the legal status, the timeline, the actors and the standards role, and needs only the inventory-duty and capacity signals as genuinely new inputs. It is reported as a quiet, secondary profile section, not a new map colour, so the structural map remains the hero and colour keeps its single meaning. It is shown as six transparent sub-scores with their sources, not a composite league-table number, which is what separates a defensible evaluation from a ranking the evidence cannot support. The limits travel with it openly: the criteria are qualitative judgements, controlled but not removed by reconciliation, and they degrade where a national strategy is too nascent to evaluate, the normal condition in this field (Paglieri et al., 2025). Reported this way, the layer adds the evaluative reading the Atlas lacks without diluting the structural reading it does well.

References

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